

ECOMMERCE SALES DATA

PROJECT OBJECTIVE

- Clean and organize raw data
- Handle missing and inconsistent values.
- Integrate multiple tables using XLOOKUP/VLOOKUP
- Calculate key business metrics such as sales, cost, profit and discount value
- Perform descriptive statistical analysis.
- Create Pivot tables for business analysis
- Develop an interactive dashboard using charts and slicers.
- Generate insights to support business decision-making.

PROJECT OBJECTIVE

This project focuses on cleaning, integrating, analysing, and visualizing an e-commerce retail dataset using Microsoft Excel. The objective is to transform raw transactional data into meaningful business insights through data cleaning, Excel formulas, Pivot Tables, descriptive statistics, and an interactive dashboard. The project helps management understand sales performance, profitability, customer behaviour, and regional trends.

PROJECT OBJECTIVE

The project using the following tables:

- Sales_Fact
- Customer_Dim
- Product_Dim
- Store_Dim
- Integrated dataset

RAW DATA

	A	B	C	D	E	F	G	H
1	Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock	
2	PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	416	
3	PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	181	
4	PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc		352	
5	P-3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	452	
6	PROD4	Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306.46	487	
7	PROD5	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	261	59	
8	PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	312.8	350	
9	PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	218.86	170	
10	PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	142	
11	PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	495.72	323	
12	PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	308	
13	PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	449.4		
14	PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	460	
15	PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	453	
16	PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson and Bowman	48.01	439	
17	PROD15	4K Home Projector	Electronics	Appliances	Thomas LLC	465.34	301	
18	P-16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256	
19	PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy and Wilson	144.53	367	
20	PROD18	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix		343	
21	PROD19	Knee Support Brace	Sports	Equipment	May-Reese	21.34	65	
22	PROD20	Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	131.51		
23	PROD21	Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	123.46	76	
24	PROD22	Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda and Rodriguez	484.62	432	
25	PROD23	Men's Formal Shirt	Clothing	Tops	Moon, Mora and Santos		360	
26	PROD24	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	166.06	340	
27	PROD25	Women's Scarf Set	Clothing	Accessories	Brown and Sons	174.02	268	
28	PROD26	Wireless Bluetooth Speaker	Electronics	Audio	Reed-Subido	223.44	242	

INCONSISTENT DATA

P-3
PROD4
PROD5
PROD6
PROD7
PROD8
PROD9
PROD10
PROD11
PROD12
PROD13
PROD14
PROD15
P-16
PROD17

CONSISTENT DATA

PROD3
PROD4
PROD5
PROD6
PROD7
PROD8
PROD9
PROD10
PROD11
PROD12
PROD13
PROD14
PROD15
PROD16
PROD17
PROD18

Descriptive Analysis

Cost		Stock		Total inventory	
Mean	254.2375107	Mean	273.4948454	Mean	70361.97044
Standard Error	11.43995937	Standard Error	11.39958015	Standard Error	225.3644756
Median	247.4009589	Median	273.4948454	Median	56946.995
Mode	358.47	Mode	273.4948454	Mode	97862.31
Standard Deviation	130.8726704	Standard Deviation	129.9504275	Standard Deviation	50789.14685
Sample Variance	17127.65587	Sample Variance	16887.11361	Sample Variance	2579537438
Kurtosis	-0.952788123	Kurtosis	-1.152941229	Kurtosis	0.491906078
Skewness	0.047919219	Skewness	-0.007691341	Skewness	0.925385707
Range	478.41	Range	439	Range	216259.36
Minimum	17.31	Minimum	57	Minimum	1112.64
Maximum	495.72	Maximum	496	Maximum	217372
Sum	25423.75107	Sum	27349.48454	Sum	7036197.044
Largest(1)	495.72	Largest(1)	496	Largest(1)	217372
Smallest(1)	17.31	Smallest(1)	57	Smallest(1)	1112.64

COST AND STOCK COLUMN MISSING VALUES:

=IF(ISBLANK(F2),AVERAGE(\$F\$2),F2)

=IF(ISBLANK(G3),AVERAGE(\$G\$2:\$G\$101),G3)

Created a manage relationships:

Status	Table	Related Lookup Table	
Active	Sales (Customer_ID)	customer (Customer_ID)	New...
Active	Sales (Product_ID)	Table1 (Product_ID)	Auto-Detect...
Active	Sales (Store_ID)	store (Store_ID)	Edit...
			Activate
			Deactivate
			Delete
			Close

Creating the chart for the category VS revenue



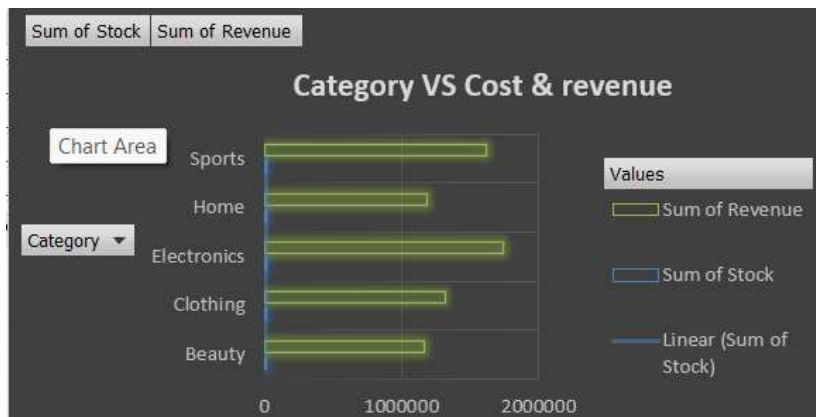
Creating the chart for the category VS stock



Creating the chart for the Payment type vs total amount



Creating the chart for the Category vs Cost and Revenue



Creating the chart for the Store vs Revenue



Dashboard:



